Report Task 1

Artificial Intelligence and Expert Systems

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1. Features

The model uses the following input features:

- x_1 (X coordinate)
- x_2 (Y coordinate)
- $\sin(x_1)$
- $\sin(x_2)$

Trigonometric inputs are to introduce nonlinearity, which helps the model learn curves a lot faster.

2. Number of hidden layers

- 1 hidden layer

3. Number of neurons in each layer

We considered two implementations, one with 5 neurons in the hidden layer and one with 4 neurons in the hidden layer.
The 5 neuron setup allowed us to minimise the test loss parameter to just 0.007, while the 4 neuron setup minimised the number of neurons while still keeping the test loss value below 0.04.

Case 1:
- Hidden layer: 4 neurons
- Output layer: 1 neuron (binary classification)

Case 2:
- Hidden layer: 5 neurons
- Output layer: 1 neuron (binary classification)

4. Activation function

- Activation function: Sigmoid

It helps the network learn smooth, curved, nonlinear decision boundaries instead of sharp, boxy ones like ReLU would produce (tanh also works pretty well but it needs 6 neurons).

5. Conclusions concerning the neural network design

This small network is simple but powerful enough to learn the spiral pattern. Because it's small and balanced, it generalizes well without needing extra regularization. Adding more layers would make it unnecessarily complex and slower to train.

6. Learning rate

The learning rate is 0.03. This value lets the network learn quickly but still converge smoothly without oscillating.

7. Regularization type

No regularization was used. Because the model is small and doesn't overfit, regularization would only make learning harder.

8. Regularization rate

Does not matter - regularization was not used.

9. Test loss value

With the 5 neuron (in the hidden layer) setup we managed to get the test loss value to just 0.007.
With the 4 neuron (in the hidden layer) setup the test loss value came down to 0.032.

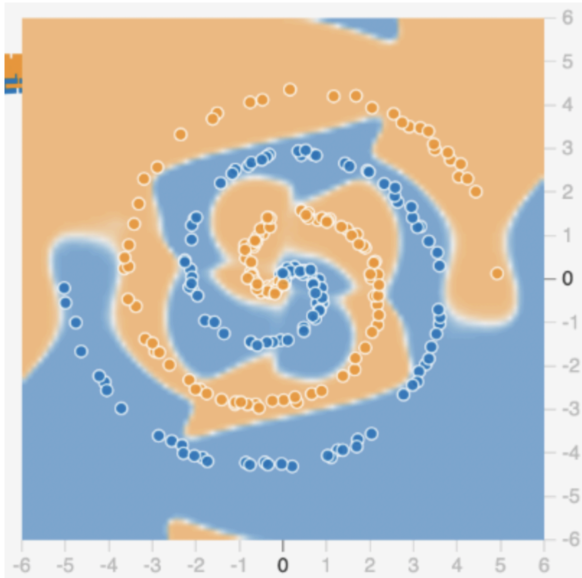
It shows that the model generalizes well and predicts accurately on unseen data.

10. Training loss value

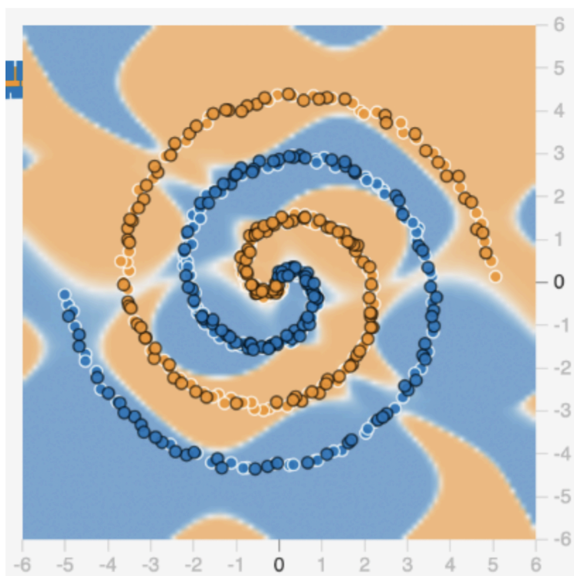
5 neuron setup: Training loss: 0.003 - lower than test loss, which is normal.
It means the network learned the training data very well without memorizing noise.
4 neuron setup: Training loss: 0.044 - higher than test loss, which is not normal.
It means we got lucky and result is effect of random fluctuations more than a consistent pattern.

11. Output image

5 neuron setup:

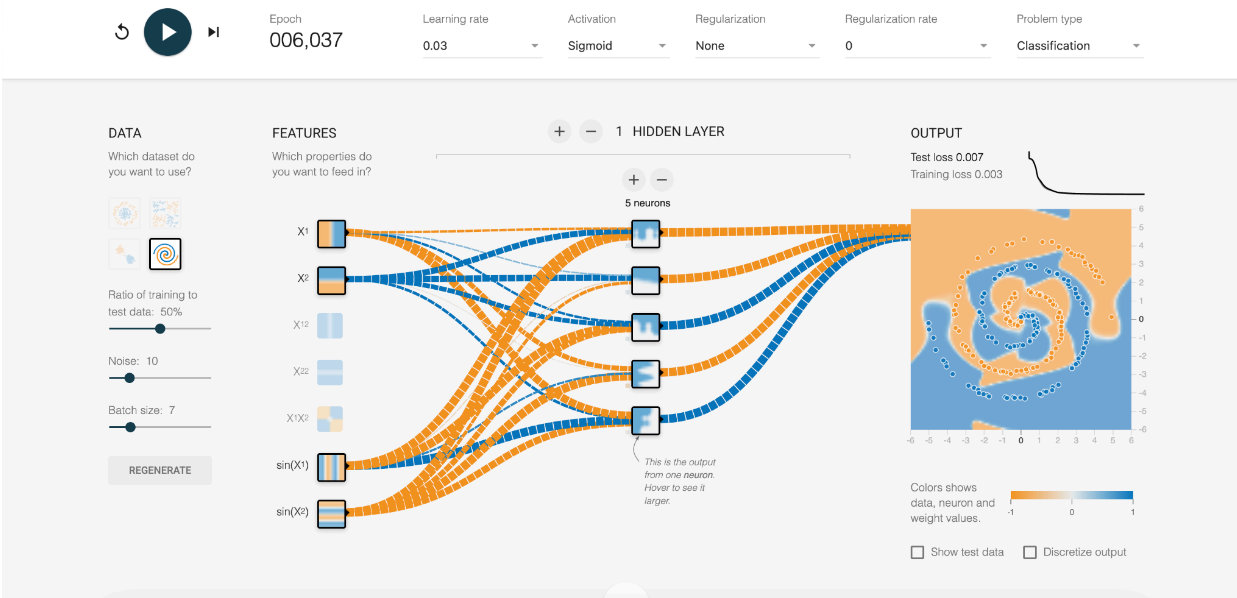


4 neuron setup:

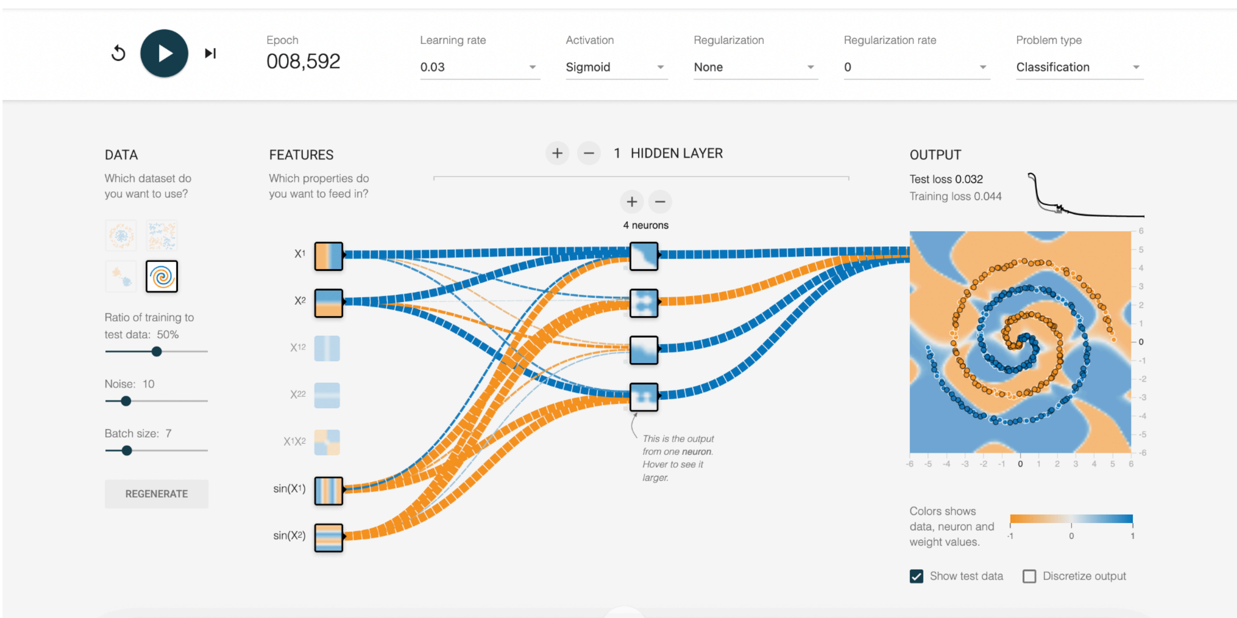


12. Screen of NN

5 Neuron setup:



4 neuron setup:



13. Conclusions

The designed neural network successfully meets the objectives of the task, demonstrating that even a small, single-hidden-layer architecture can effectively model nonlinear relationships such as the spiral pattern.